PyMap - User Manual

1. Introduction

Pymap is a custom, command-line network reconnaissance and diagnostic tool designed to simulate the core functionalities of Nmap. It allows users to perform network mapping through multithreaded host discovery, OS fingerprinting, TCP port scanning, and service version detection.

2. Prerequisites & Installation

- **Operating System:** Kali Linux, Ubuntu, or Windows (must be run as Administrator/root).
- **Environment:** Python 3.x
- **Dependencies:** The tool requires the `scapy` library for raw packet crafting.
- **Installation Command:** `pip install scapy`

3. Usage Instructions

Pymap is executed via the command-line interface. Because the tool crafts raw ICMP packets for host discovery and OS fingerprinting, it strictly requires elevated system privileges.

Base Syntax:

```
sudo python3 pymap.py -t <TARGET_IP_OR_SUBNET> [OPTIONS]
```

Arguments Reference

Argument	Full Flag	Required	Description
`-t`	`--target`	**Yes**	Target IP address or CIDR notation (e.g., `192.168.56.200` or `192.168.56.0/24`).
`-p`	`--ports`	No	Port range to scan formatted as START-END. Defaults to `1-1000`.

-s	--sweep-only	No	Skips port scanning and only performs a multithreaded ping sweep to discover live hosts.

4. Execution Examples

Host Discovery (Ping Sweep)

To map an entire subnet and discover live hosts alongside their guessed operating systems rapidly:

sudo python3 pymap.py -t 192.168.56.0/24 -s

Full Deep Scan (Discovery, OS Guessing, & Port Scanning)

To perform host discovery, followed by a multithreaded TCP port scan and service banner grab on a specific machine:

sudo python3 pymap.py -t 192.168.56.200 -p 1-1000

Custom Port Range Scan

To target specific ports on a known host:

sudo python3 pymap.py -t 192.168.56.200 -p 80-500